



CI / CD

Introduction to DevOps

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What is CI?

Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily - leading to multiple integrations per day. Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible. Many teams find that this approach leads to significantly reduced integration problems and allows a team to develop cohesive software more rapidly.

- Martin Fowler (2006)

-> CI means to automatically verify that the update to the software source code is working as intended

What is CD?

Continuous Delivery is a software development discipline where you build software in such a way that the software can be released to production at any time. You achieve continuous delivery by continuously integrating the software done by the development team, building executables, and running automated tests on those executables to detect problems. Furthermore you push the executables into increasingly production-like environments to ensure the software will work in production. To do this you use a Deployment Pipeline.

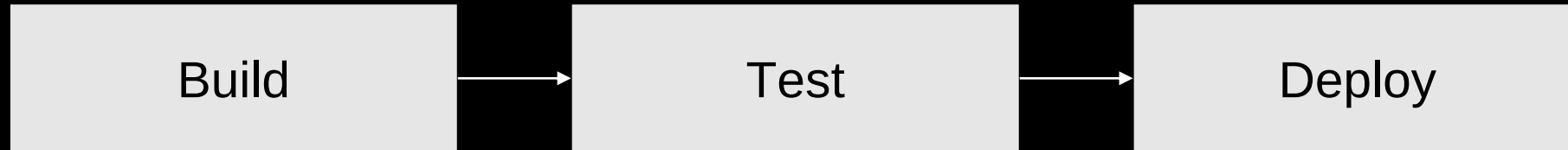
- Martin Fowler (2013)

-> CD means to automatically publish the software when the source code is updated

CI / CD & DevOps

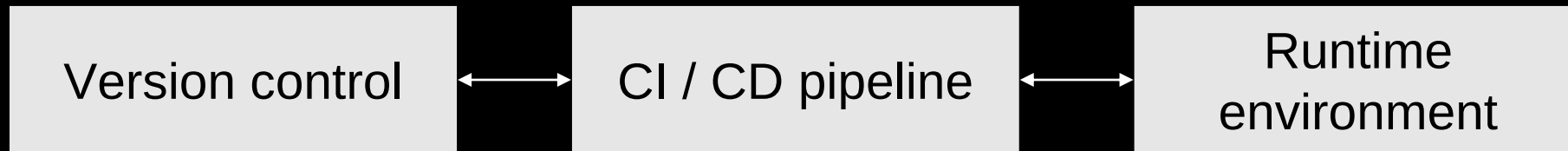
- DevOps is a set of practices that enhance the co-operation between the Development & Operations of a software product
- Because CI and CD strive to automatize software integration and delivery, it can be seen as a way of practicing DevOps
- CI / CD can be implemented concretely with pipelines - the term pipeline is used for an automatic process that integrates and delivers the software

CI / CD pipeline



- The above is an example of the steps that a CI / CD pipeline can perform
- The pipeline can be run automatically when a certain event occurs in the version control
 - For example, the pipeline can be triggered when the main branch of the version control is updated

Implementing a CI / CD pipeline in practice



- The CI / CD pipeline needs to interact with:
 - The version control system (in order to detect when the software is updated)
 - The runtime environment of the software (in order to deploy the software automatically)
- Ready-made integrations that are available on some tools make the implementation of CI / CD pipeline more streamlined
 - The available integrations are a major factor when deciding the tool with which the pipeline is to be implemented

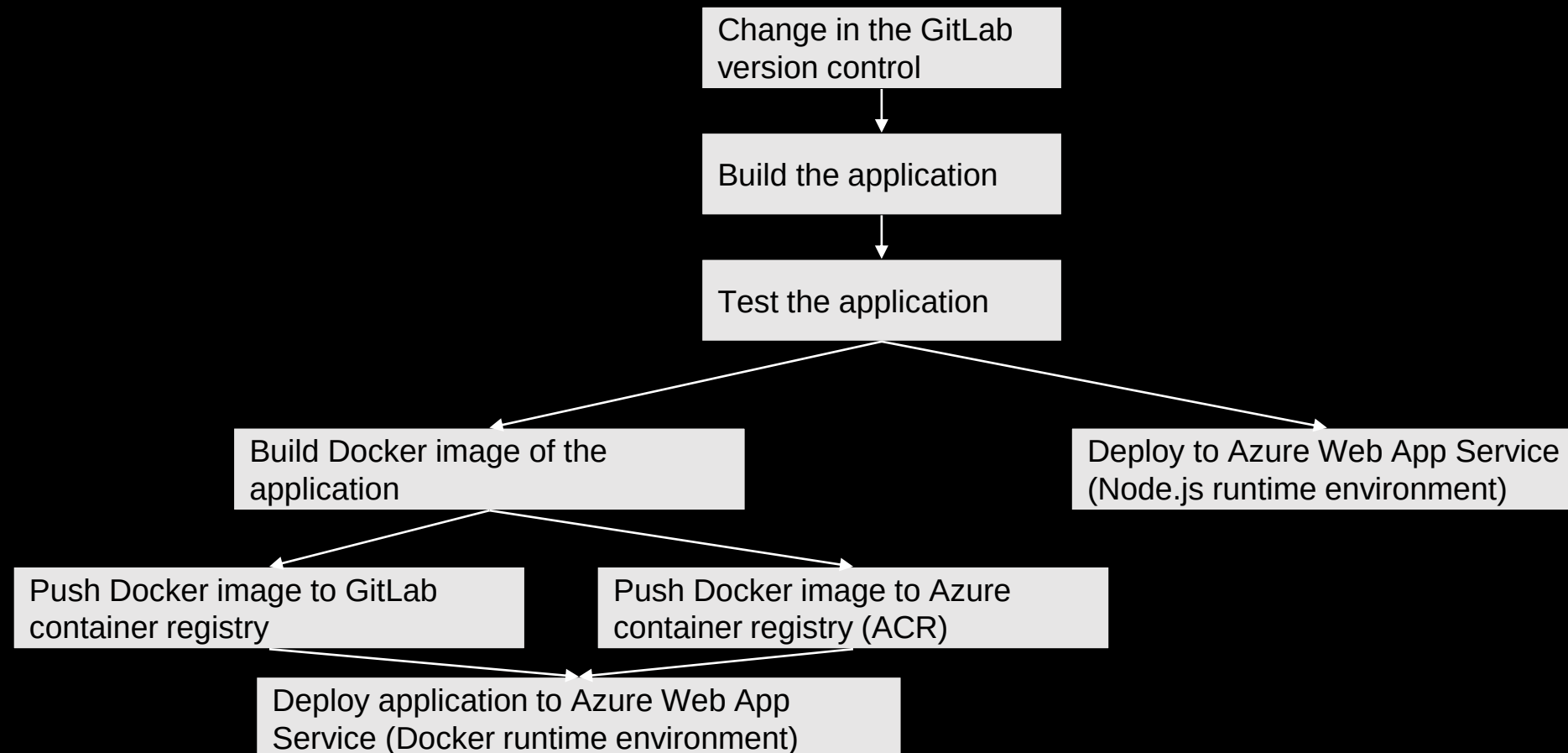
CI / CD tools and integrations

- There are many tools that can be used to implement CI / CD pipelines
 - GitHub Actions
 - GitLab CI / CD
 - CircleCI
 - Jenkins
 - others...
- Examples of direct integrations:
 - GitHub <-> GitHub Actions <-> Microsoft Azure services
 - GitLab <-> GitLab CI / CD
- Even when direct integrations are not available, the services can usually interact with each other using for example API calls, webhooks, etc.

Concrete CI / CD implementation with GitLab and Microsoft Azure

- Let's assume that there is a Node.js-based web application on GitLab that is to be deployed automatically to Microsoft Azure
 - In this case, GitLab CI / CD is used as the tool to implement the CI / CD pipeline (other options like Jenkins could also be used, but in this case GitLab is the most efficient due to the CI / CD solution being built-in into it)
- Option A: Deploy the application via Web App Service using the Node.js runtime environment
- Option B: Containerize and deploy the application via Web App Service using the Docker runtime environment
 - Further options: use either Microsoft Azure or GitLab as the container registry, where the container image of the software is pushed

Visualization of the possible CI / CD pipelines



References

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THANK YOU!

